

# Dual-Database Retrieval-Augmented Generation (RAG) Architecture & Workflow

Spring Boot 4.1.1 • Spring AI 2.0.1 • MongoDB 7.0 • Qdrant HNSW • Ollama Llama 3.3 (70B) & Mistral Large • SAP Fiori Horizon UI

📄 Document Upload & Processing Pipeline

## 1. PDF / Doc Upload

- Drag & drop file
- Auto-extract BibTeX
- Extract full text
- SHA-256 Checksum

DocumentRecord

## 2. Semantic Chunking

- Chunk size: ~500 chars
- 100 char sliding overlap
- Sentence boundary split
- Preserves context flows

Chunks: [C0, C1, ... Cn]

## 3. nomic-embed-text

- Local Ollama Embedding Model
- 768-dimensional float32
- Unit L2 normalized
- High-throughput batching

$v \in \mathbb{R}^{768} \text{ (} \|v\| = 1.0 \text{)}$

📄 Document Metadata & Asset Storage

- Stores full BSON, BibTeX, versions, PDF asset
- Caches dual Llama & Mistral summaries
- ACID Guarantees • B-Tree Index • Port 27017

## Qdrant Vector DB (Vector Index)

- Stores 768-D dense vectors + chunk payload
- Hierarchical Navigable Small World (HNSW)
- < 3ms Cosine Distance ANN • Port 6333/6334

## ⚙️ Concurrent Summarization

1. Llama 3.3 (70B Instruct):  
Analytical synthesis • Methodologies
2. Mistral Large (2411):  
Executive brief • Key findings

Concurrent Async Threads  
Duration tracked & rendered in Object Page

🔍 Query Processing & Retrieval Workflow

📄 User Input & Context

Natural Language Question:

"What are the primary findings of this paper?"

- Emitted from Object Page
- POST /api/v1/documents/:id/chat
- Passes prior chat history

RestBackendAdapter

Keycloak Bearer JWT Header  
nginx Ingress → Spring Boot

🔍 Query Processing & Retrieval Workflow

## 1. Query Vectorization:

$v_{\text{query}} = \text{Embed}(\text{question})$

## 2. Filtered Cosine Distance:

$\text{doc\_id} = \text{currentDocument}$   
Threshold:  $\cos(\theta) > 0.65$

## 3. Top-K Extraction (K=3):

- Chunk #2 (Score: 0.94) • Page 4  
"Multi-head attention allows..."
- Chunk #0 (Score: 0.88) • Page 1  
"We propose the Transformer..."
- Chunk #7 (Score: 0.73) • Page 9

📄 Retrieved Context & Prompts

## [SYSTEM PROMPT]

You are Spring AI Assistant powered by Llama 3.3. Ground responses strictly in the retrieved passages with citations.

## [RETRIEVED PASSAGES]

[Excerpt 1] Multi-Head Attention ...  
[Excerpt 2] We propose the Transformer ...

## [RECENT CHAT TURNS]

User: Tell me about parallelization.  
Assistant: Unlike RNNs, Transformer ...

## [USER QUESTION]

"What are the primary findings?"

Zero-Hallucination Guardrail Active

📄 Streaming Response & Audit Trail

## Streaming Synthesized Response (Markdown):

"The primary findings demonstrate that the **Transformer** replaces recurrent layers with **Multi-Head Attention** [Excerpt 1]. On WMT 2014 translation benchmarks, the big model achieves 28.4 BLEU, training in 3.5 days on 8 GPUs [Excerpt 2]."

✓ Model: Llama 3.3 (70B Instruct) via Spring AI • Latency: 1.8s  
Directly verifiable in SAP UI5 Object Page with interactive excerpt cards

📄 Audit Trail & Metadata

- Chunk 2 (Relevance: 94.2%) – "Multi-Head Attention ..."
- Chunk 0 (Relevance: 88.0%) – "Attention Is All You Need ..."
- Full audit trail preserved in MongoDB document version logs